B. Comparison String

time limit per test 2 seconds
memory limit per test 512 megabytes
input standard input
output standard output

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You are given a string s of length n, where each character is either < or >.

An array a consisting of $n+1$ elements is compatible with the string s if, for every i from 1 to n, the character s_i represents the result of comparing a_i and a_{i+1}, i. e.:

- s_i is < if and only if a_i lt a_{i+1};
- s_i is > if and only if a_i gt a_{i+1}.

For example, the array [1, 2, 5, 4, 2] is compatible with the string <<>>. There are other arrays with are compatible with that string, for example, [13, 37, 42, 37, 13].

The **cost** of the array is the number of different elements in it. For example, the cost of [1, 2, 5, 4, 2] is 4; the cost of [13, 37, 42, 37, 13] is 3.

You have to calculate the minimum cost among all arrays which are compatible with the given string s.

Input

The first line contains one integer t ($1 \le t \le 500$) — the number of test cases.

Each test case consists of two lines:

- the first line contains one integer n ($1 \le n \le 100$);
- the second line contains the string s, consisting of n characters. Each character of s is either < or >.

Output

For each test case, print one integer — the minimum cost among all arrays which are compatible with the given string s.

Example

Input
4 4 <<>> 4 >><< 5 >>>>> 7 <><><><

Output
3 3 6 2

Note

In the first test case of the example, the array can be [13, 37, 42, 37, 13].

In the second test case of the example, the array can be [42, 37, 13, 37, 42].